Homework #3: Vector Embeddings and RAG

Write-Up Report

Name: Asad Khan

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Assignment: Homework #3 - Vector Embeddings and RAG

Collaborators (coding and report): None. GenAI disclosure: Claude (Anthropic) was used as a coding assistant, to help with the manual scoring, and to draft this report.

Files submitted: `hw3_embeddings_AsadKhan.ipynb` (completed starter notebook, renamed) and this report.

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1. Introduction and Methodology

1.1 RAG and the goal of the experiment

Retrieval-Augmented Generation (RAG) answers a question in two steps. A **retriever** embeds the question and returns the k text chunks whose embeddings are closest to it in a vector store. A **generator** (here gpt-4o-mini) is then asked to answer using only those chunks. The LLM can only be as good as the evidence it is handed: if the retriever never surfaces a passage, the generator cannot use it, however capable it is. The goal of this homework was to measure that dependency on a realistic corpus. Experiment A compares two sentence-transformer embedders, `all-MiniLM-L6-v2` (384 dimensions, 6 layers) and `all-mpnet-base-v2` (768 dimensions, 12 layers). Experiment B varies chunk size, chunk overlap and k for MiniLM. Five questions of increasing difficulty were scored by hand.

1.2 Corpus

Table 1. The four documents after loading and chunking (default chunk_size 1000, overlap 200).

Short name	Document	Pages	Characters	Chunks
Blueprint	White House Blueprint for an AI Bill of Rights (2022)	73	220,968	299
EO 14110	Executive Order 14110 on Safe, Secure, and Trustworthy AI (2023), Federal Register print	36	145,506	194
NIST AI RMF	NIST AI 100-1, AI Risk Management Framework 1.0	48	106,444	148
EU AI Act	EU Artificial Intelligence Act, COM(2021) 206 proposal	108	314,714	424
Total		**265**	**787,632**	**1,065**

Inspecting the files showed something the assignment description does not mention. The "EU AI Act" file is not a short summary of key principles: it is the full 108-page 2021 Commission proposal (explanatory memorandum, 89 recitals, 85 articles and the financial statement), *without* its annexes. It alone supplies 40% of all chunks, so with a fixed k it has the most chances to crowd out the smaller

documents. Its legal register ("this Regulation", "the Union", "providers") also differs sharply from the US documents, which turned out to matter for retrieval.

1.3 Pipeline and text-splitting strategy

1. **Loading (TODO 1).** `PyPDFLoader` yields one `Document` per page. Each page is tagged with metadata (`source_title`, a short name such as "EU AI Act", and a 1-based `page_number`), so every chunk can be traced back to its document and page. The loader uses the local `data/` copy when present and otherwise downloads from the course GitHub, so the notebook also runs on Colab.
2. **Cleaning.** PDF extraction leaves artifacts that waste embedding capacity. I removed the Federal Register running header that repeats on every EO page, the Blueprint's "SECTION TITLE" stamps and the EU "EN n EN" footers. I re-joined words hyphenated across line breaks ("respon-sible") and collapsed visual line breaks into running text. Some artifacts cannot be repaired by regex. The most important is that superscripts are flattened, so 10^{26} appears in the text as "10 26". I left these in deliberately, as a test of the generator.
3. **Splitting (TODO 2).** `RecursiveCharacterTextSplitter` with chunk_size 1000, overlap 200 and the separator list `["\n\n", "\n", ". ", "; ", ", ", " ", ""]` with `keep_separator="end"`. Once line breaks are collapsed, the splitter cuts at sentence ends first, then clause boundaries, and only splits inside a sentence as a last resort. `"end"` keeps the period with its sentence instead of starting the next chunk with ". ". Chunks shorter than 100 characters (cover pages, page-number fragments) are dropped. Mean chunk length is 768 characters, about 159 word-piece tokens.
4. **Embedding and indexing (TODO 3).** `HuggingFaceEmbeddings` with `normalize_embeddings=True` and a Chroma collection using cosine distance (`hnsw:space = cosine`). Each model gets its own persistent database, and the collection is deleted before it is rebuilt, so re-running the notebook never inserts duplicates.
5. **Generation (TODO 5).** The top-k chunks are formatted as labeled excerpts ("[Excerpt 2] (EO 14110, p. 7)") and sent through a LangChain chain (`PromptTemplate | ChatOpenAI | StrOutputParser`) with temperature 0. The prompt tells the model to use only the context, cite the excerpt behind each claim, treat each document separately and then synthesize, and say explicitly when a document or fact is missing. Answers are cached on disk keyed by a hash of (model, context, question), so re-running the notebook reproduces exactly the answers I scored.

Rationale for a multi-document corpus. I split per page rather than per document. The cost is that a chunk can never span a page break. The benefit is that every chunk carries an exact page citation, which the generator uses and which made manual verification fast. 1000 characters, about one legal provision or one paragraph of prose, is long enough to hold a complete rule ("both of the following conditions are fulfilled: (a) ... (b) ..."). It is also short enough that its embedding is not an average of several unrelated topics, and well under MiniLM's 256-token input limit (only 3.2% of default chunks exceed it). The 200-character overlap protects a sentence that straddles a cut. I did *not* prepend the document title to each chunk's text. That would help any query that names a document, but it would

also hide exactly the difference between embedders that Experiment A is meant to measure. Section 4 returns to this choice.

1.4 Queries (TODO 4)

I wrote the questions myself after reading the relevant sections, aiming for questions that a single lucky chunk cannot answer. Each has several required parts, and each part is located in a known place, so retrieval can be checked chunk by chunk.

Table 2. The five test queries.

#	Query	Type, location of the answer and why it is hard
Q1	*Under the Blueprint for an AI Bill of Rights' 'Human Alternatives, Consideration, and Fallback' principle, when should people be able to opt out of an automated system, what must a fallback and escalation process provide, and which sensitive domains call for extra human oversight?*	Single document (Blueprint, pp. 7, 46-49). Three parts. The sensitive-domain list (criminal justice, employment, education, health) sits at the *end* of the principle statement, after the opt-out and fallback text.
Q2	*Under Executive Order 14110, what must companies developing dual-use foundation models report to the federal government, and what computing-power thresholds trigger these reporting requirements for models and for computing clusters?*	Single, different document (EO Sec. 4.2, p. 7). Needs the reporting list (A)-(C) and three numeric thresholds (10^{26} operations; 10^{23} for biological sequence data; clusters with 100 Gbit/s networking and 10^{20} operations per second), which appear in the text as "10 26".
Q3	*What mechanisms do the Blueprint for an AI Bill of Rights, the EU AI Act, and the NIST AI RMF each propose to keep humans in control of AI systems (human oversight, intervention, or fallback), and what shared view of human accountability emerges across them?*	Multi-document synthesis. Search and combine: find the human-control mechanism in three documents (Blueprint fallback principle; EU Art. 14 "Human oversight"; NIST GOVERN and MANAGE 2.4 "supersede, disengage, or deactivate"). Thematic synthesis: state the common view. The three documents use different vocabulary for the same idea.
Q4	*How does the NIST AI RMF's voluntary approach to managing AI risk differ from the EU AI Act's legally binding obligations for high-risk AI systems, in terms of who must comply, how risk is assessed, and what happens before a system can be deployed?*	Comparative, with three axes of contrast. Needs NIST's "voluntary, non-prescriptive" design and EU's ex-ante conformity assessment, registration and provider obligations.
Q5	*How does the EU AI Act proposal define an 'AI system', and under Article 6, what conditions make an AI system 'high-risk'?*	Definition and scope inside the longest document. Two exact passages about 6 pages apart: Art. 3(1) on p. 40 and Art. 6(1)-(2) on p. 46. The explanatory memorandum and recitals paraphrase both passages and compete with the real text.

A pilot finding. My first Q5 asked which "techniques and approaches listed in Annex I" fall under the AI-system definition. Every run failed to retrieve Annex I. Searching the extracted text showed why:

the annexes are a separate EU document and are not in this PDF. Both models' answers stated that the Annex I list was not in the context rather than inventing one (mpnet even retrieved the Art. 3(1) definition, which points to Annex I). This was a good sign for the prompt's honesty rule, but an unanswerable question cannot measure retrieval quality, so I replaced it with the Article 6 question above.

1.5 Evaluation

The starter's `evaluate_results` prompts with `input()`. I kept that path (set `INTERACTIVE_EVAL = True`) but recorded the 55 judgments (11 configurations x 5 queries) in a `MANUAL_SCORES` table, so the notebook runs end to end and shows every chunk and score. Every score was assigned after reading every retrieved chunk in full and checking the answer against the source text. I applied the rubric as follows:

- **Retrieval:** 3 = the chunks are on-topic, from the right document(s), and contain the evidence for every part of the question; 2 = relevant material but a required part or a required document is missing, or more than a few chunks are noise; 1 = mostly generic or off-topic text.
- **Answer:** 3 = every part answered correctly from the context; 2 = partially correct (a part missing, or a minor misattribution); 1 = key parts missing (for example a comparison declined because one side is absent) or a fabricated claim.

As an objective cross-check I also computed **document coverage**: the fraction of the documents a question needs (Q1, Q2, Q5: one; Q4: two; Q3: three) that appear among the retrieved chunks.

2. Results and Analysis: Experiment A (embedding models)

Table 3. Experiment A, default parameters (chunk_size 1000, overlap 200, k 4). Averages over the five queries.

Model	Dim.	Avg retrieval (1-3)	Avg answer (1-3)	Doc coverage	Time to embed 1,065 chunks*
all-MiniLM-L6-v2	384	2.20	1.60	0.77	2.7 s
all-mpnet-base-v2	768	2.20	**2.00**	**0.93**	21.2 s

* Apple-silicon laptop (MPS), batch size 32, measured once after warm-up.

Table 4. Per-query scores (retrieval / answer) and the sources of the four retrieved chunks.

Q	MiniLM	MiniLM sources	mpnet	mpnet sources
Q1	2 / 2	Blueprint p.7, p.47, p.47, p.46	2 / 2	Blueprint p.7, p.8, p.8, p.2
Q2	3 / 2	EO p.7, p.7, p.7, p.13	2 / 2	EO p.7, p.13, p.13, p.7
Q3	2 / 1	Blueprint p.7, p.4, p.8; EO p.28	2 / 2	EU p.52, Blueprint p.9, p.7, EU p.10
Q4	2 / 1	NIST p.8, p.37, p.8, p.7	3 / 2	NIST p.47; EU p.21, p.26, p.15
Q5	2 / 2	EU p.27, p.14, p.10, p.46	2 / 2	EU p.46, p.78, p.80, p.14

The average retrieval scores tie, but the averages hide what matters. MiniLM's chunks were usually on-topic *within one document*, and it never reached a second document when a question required one. mpnet sometimes pulled a weaker chunk, but it reached the EU AI Act on both cross-document questions. That is why its answer score is higher. Coverage shows the same pattern: 0.77 vs 0.93.

2.1 Qualitative comparison, question by question (best = mpnet, worst = MiniLM)

Q1 (Blueprint, single document): same score, different failure.

MiniLM's four chunks were all about the right principle, but only two pieces of text were distinct. The principle statement is printed twice in the Blueprint (the summary on p. 7 and the principle page on p. 46), and both copies ranked in the top 4, along with two paragraphs of "Why this principle is important" from p. 47. More importantly, the 1000-character cut on p. 7 falls exactly after "...should not impose an unreasonable burden on the public." The very next sentence, "Automated systems with an intended use within sensitive domains, including ... criminal justice, employment, education, and health...", went into the following chunk, which did not rank. The generator handled this correctly: it answered the opt-out and fallback parts accurately and said that "the excerpts do not specify which sensitive domains call for extra human oversight." mpnet retrieved that second chunk, with the domain list, at rank 1, but lost the opt-out sentence. Its other three chunks were generic Blueprint framing (the "overlapping set of backstops" passage, the two-part scope test, and the legal disclaimer on p. 2). Its generator then stitched the pieces together wrongly: "people should be able to opt out of an automated system *when it is used in sensitive domains*". The Blueprint never says that. Both runs scored 2/2, but MiniLM's answer is incomplete and honest, while mpnet's is incomplete *and* contains a misattribution.

Q2 (Executive Order, single document): MiniLM slightly better at retrieval.

Both models found Sec. 4.2 on p. 7. MiniLM spent three of its four slots there: the reporting list (A)-(B), the threshold paragraph, and the "technical conditions" paragraph. mpnet spent two slots on Sec. 4.6 (p. 13, models with widely available weights). That section repeats the phrase "dual-use foundation models" six times but is about a different policy question. The answers were identical in content and correct on all three thresholds. The generator converted the flattened "10 26" and "10 20" into 10^{26} and 10^{20}, so it repaired a PDF artifact that would mislead a keyword system. Both answers missed reporting item (C), the results of red-team testing, because the default chunk ends at item (B) and the next chunk did not rank.

Q3 (multi-document synthesis): the largest difference.

MiniLM retrieved three Blueprint chunks and one EO chunk, and **no chunk from the EU AI Act or the NIST AI RMF**. Only one Blueprint chunk (p. 7, the human-fallback text) was actually about human control. The other two were generic introductions (p. 4 "About this framework" and p. 8), which ranked because they repeat the phrase "Blueprint for an AI Bill of Rights", one of the strings in the query. 30 Blueprint chunks contain the document's full title. The EO chunk (p. 28) mentions "practices derived from OSTP's Blueprint for an AI Bill of Rights and the NIST AI Risk Management Framework ... granting human consideration and remedies for adverse decisions". The generator used

it to describe NIST's mechanisms *second-hand*, through the EO, and correctly said the EU AI Act was missing. mpnet retrieved what is arguably the single most relevant passage in the corpus: EU Article 14 "Human oversight" (p. 52, "High-risk AI systems shall be designed and developed ... [so] that they can be effectively overseen by natural persons"). It also retrieved EU p. 10 (human oversight listed among the mandatory requirements) and the same Blueprint p. 7 chunk. That is two of the three documents, with real content, and the answer covered both, synthesized them, and flagged NIST as missing (score 2 vs 1). Neither model found NIST's own human-control language. NIST rarely uses the word "oversight": it talks about "human-AI teaming", "AI actors" and mechanisms to "supersede, disengage, or deactivate AI systems" (MANAGE 2.4). This is the vocabulary-mismatch problem that dense retrieval is supposed to solve, and a 384- or 768-dimension general-purpose encoder only partly solves it.

Q4 (comparative): MiniLM failed to retrieve a required document.

This is the clearest case of a model failing to retrieve a crucial document. All four MiniLM chunks came from NIST: the RMF overview on p. 8 (twice), the MANAGE table on p. 37, and the outline on p. 7. 44 of the 148 NIST chunks begin with the running header "NIST AI 100-1 AI RMF 1.0", so any query that names "NIST AI RMF" gets a lexical boost from text that carries no content. With no EU text, the generator declined the comparison ("The context provided does not include any information about the EU AI Act"), which is the correct behavior but scores 1. mpnet retrieved NIST Appendix D (p. 47, the RMF should "be risk-based, resource-efficient, pro-innovation, and voluntary") plus three EU chunks, including p. 15 on the "ex-ante and ex-post compliance and enforcement mechanisms" and third-party conformity assessment. It produced a clean three-axis comparison. It still scored 2, not 3: for "who must comply" under the EU Act, it relied on recital 12 (p. 21) and wrote that the regulation "applies to Union institutions, offices, bodies, and agencies". That recital is about extending the regulation to EU institutions, not about who is primarily bound (providers and users).

Q5 (definition and scope): mpnet finds the article, MiniLM finds paraphrases of it.

Neither model retrieved the Art. 3(1) definition on p. 40, so both answers correctly said the definition was not in the context. (In the pilot, when the question said "artificial intelligence system" instead of "AI system", mpnet did retrieve Art. 3(1) at rank 1. The shorter wording plus "Article 6" pulled every top result toward classification.) The difference is in the Article 6 half. mpnet's rank-1 chunk is the actual text of Art. 6(1) with both conditions (a) and (b), and its answer quotes them exactly. MiniLM's chunks were recital 32 (p. 27), the explanatory memorandum (p. 14), the policy-options page (p. 10) and the *second* half of p. 46 (Art. 6(2) and Art. 7). Its answer says that under Article 6 a system is high-risk "if, based on its intended purpose, it poses a high risk of harm to health and safety or fundamental rights ... severity ... probability". That wording comes from the recital and from Art. 7 (the criteria for *amending* Annex III), not from Art. 6. Both runs scored 2/2, but mpnet's answer is precisely correct for the half it answers, while MiniLM's is a plausible paraphrase of the wrong provision.

2.2 Correlation between retrieval quality and answer quality

Across all 55 scored runs (both experiments), the Pearson correlation between retrieval and answer scores is **0.64**. More telling, the answer score **never exceeded** the retrieval score (55 of 55 runs have answer <= retrieval). The joint counts were (retrieval, answer) = (2,2) x 20, (2,1) x 18, (3,2) x 7, (1,1) x 6, (3,3) x 4. Good retrieval is necessary but not sufficient: the 25 runs that lost a point between retrieval and answer are the interesting ones.

Generator failures (good retrieval, weaker answer):

- **Chunk size 2000, Q5.** The context contained Art. 6(1) and Art. 6(2) ("AI systems referred to in Annex III shall also be considered high-risk"). The answer reported only the two conditions of paragraph 1 and silently dropped paragraph 2, the second route to high-risk status.
- **Overlap 400, Q1.** The domain list was not in the context. Instead of saying so, as the default run did, the model invented an answer: "sensitive domains ... include situations where rights, opportunities, or access are meaningfully at stake, as well as cases where the automated system may reinforce bias". That is a rewording of the "why this matters" paragraph presented as the requested list.
- **k = 2 and k = 12, Q5.** Despite the instruction to use only the context, both runs began "The EU AI Act proposal defines an 'AI system' as a system that is capable of performing tasks that typically require human intelligence" (k = 2), or "...any system that uses artificial intelligence techniques..." (k = 12), and then admitted that the definition was not in the excerpts. These are textbook definitions from the model's pre-training, not the Act's text (which says "software that is developed with one or more of the techniques and approaches listed in Annex I..."). Both received answer score 1.
- **Citation errors.** With chunk size 2000 on Q3, the model cited EO p. 28 content as "[Blueprint, p. 28]". With k = 8 on Q1, it cited the p. 7 chunk as "Excerpt 5, p. 49". The facts were right but the provenance was wrong, which matters in a policy setting.

Generator successes (imperfect retrieval, good answer):

- **Q2, all runs.** The threshold chunk is corrupted by flattened superscripts ("greater than 10 26 integer or floating-point operations", "greater than 1023"). gpt-4o-mini rendered all three thresholds correctly as 10^{26}, 10^{23} and 10^{20}.
- **k = 12, Q4.** The only EU chunk was recital 32, at rank 11 of 12, below ten NIST chunks. The generator still found it and built a correct three-axis comparison. It combined NIST p. 13 ("development and deployment should cease in a safe manner until risks can be sufficiently managed") and p. 12 ("does not prescribe risk tolerance") with the recital's third-party conformity assessment. One relevant chunk at the bottom of a long context was enough.
- **mpnet Q3.** With two of three documents, the model wrote a correct per-document account plus a reasonable synthesis (the EU Act builds oversight into the design of high-risk systems, the Blueprint stresses accessible fallback and operator training, and both treat human accountability as essential in high-stakes areas). It also explicitly scoped the synthesis to the evidence it had.

2.3 Which model performed best, and why

all-mpnet-base-v2 performed best overall on this corpus: higher answer score (2.00 vs 1.60), higher document coverage (0.93 vs 0.77), and it was the only model to reach the EU AI Act on the two cross-document questions (Q3 and Q4) and the exact Article 6 text on Q5. MiniLM won only Q2, a single-document question with strong lexical overlap between query and source.

I think three factors explain this. (1) **Capacity for paraphrase.** mpnet has twice the layers and twice the embedding width. The questions most often describe a concept in words different from the source ("keep humans in control" vs "Human oversight"; "what happens before a system can be deployed" vs "ex-ante conformity assessment"), and that is where a larger encoder helps. (2) **MiniLM is more swayed by surface strings.** Its retrievals were dominated by chunks that repeat the document *names* in the query ("Blueprint for an AI Bill of Rights", "NIST AI 100-1 AI RMF 1.0" headers). In a corpus where three of four documents name themselves constantly and the EU proposal calls itself "this Regulation" (only 2 of 424 EU chunks contain "Artificial Intelligence Act"), that bias systematically starves the EU document. (3) **Longer context window.** mpnet reads 384 tokens versus MiniLM's 256, which matters little at chunk size 1000 but a lot for longer chunks (Section 3). The price is about 8x slower embedding (21.2 s vs 2.7 s for this corpus) and twice the storage per vector. That is negligible here, but it would matter for millions of chunks or for embedding queries on a CPU at serving time.

3. Results and Analysis: Experiment B (chunk size, overlap, k with MiniLM)

Design. Starting from the default configuration (1000 / 200 / 4, the MiniLM run of Experiment A), I changed one parameter at a time and re-ran all five queries, rebuilding the index for every chunking change. Values: chunk_size in {250, 500, 1000, 2000} (overlap 200, lowered to 50 for size 250 because a 200-character overlap on a 250-character chunk would make neighbours 80% identical); chunk_overlap in {0, 100, 200, 400} (size 1000); k in {2, 4, 8, 12}. Every one of the 45 runs was scored with the same rubric.

Table 5. Experiment B results (all-MiniLM-L6-v2). Per-query entries are retrieval / answer.

Configuration	Chunks	Q1	Q2	Q3	Q4	Q5	Avg ret.	Avg ans.	Cov.
size 250 (overlap 50)	3,922	2/1	2/1	1/1	1/1	1/1	1.40	1.00	0.77
size 500	2,334	2/2	2/2	1/1	2/1	1/1	1.60	1.40	0.77
default (1000 / 200 / 4)	1,065	2/2	3/2	2/1	2/1	2/2	2.20	1.60	0.77
size 2000	546	3/3	3/3	1/1	2/1	2/2	2.20	2.00	0.77
overlap 0	1,009	2/2	3/2	2/1	2/1	2/2	2.20	1.60	0.77
overlap 100	1,014	2/2	3/2	2/1	2/1	2/2	2.20	1.60	0.77
overlap 400	1,221	2/2	3/2	2/1	2/1	2/2	2.20	1.60	0.77
k = 2	1,065	2/2	3/2	2/1	2/1	2/1	2.20	1.40	0.77
k = 8	1,065	3/3	3/2	2/1	2/1	2/2	2.40	1.80	0.77
k = 12	1,065	3/3	2/2	2/2	2/2	2/1	2.20	2.00	0.93

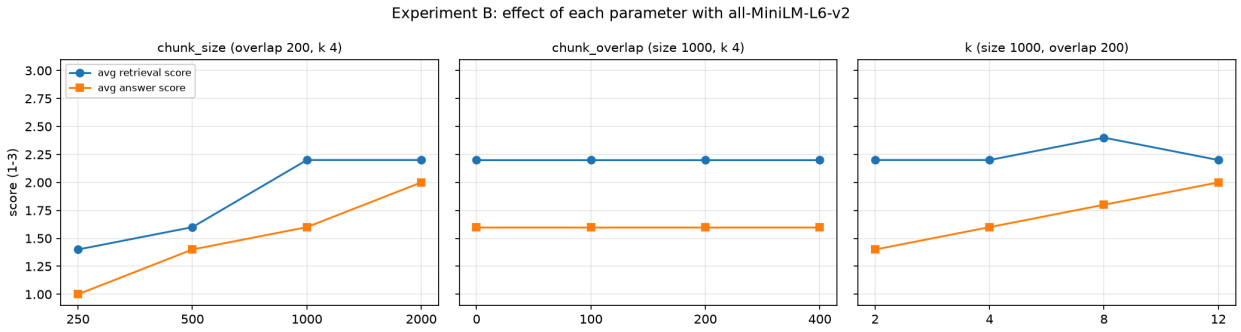


Figure 1. Average retrieval and answer score as each parameter varies (others at default).

Chunk size had the largest effect, in both directions.

Small chunks (250, 500) were clearly worse. At 250 characters a chunk is one or two sentences (median 38 tokens). For Q3, MiniLM returned four Blueprint fragments whose only content is the document title ("The Blueprint for an AI Bill of Rights is meant to assist governments..."). For Q5, it returned four fragments of procedural text that happen to contain "high-risk AI systems" ("For high-risk AI systems referred to in point 1(a) of Annex III, the measures referred to in paragraph 3 shall be such as to ensure that, in addition,"). Short chunks make the vector a fingerprint of a few words, so retrieval matches surface phrases, and even a correct hit carries too little context to answer from. On Q2, the size-250 retrieval contained only the 10^{26} part of the threshold sentence, and the answer reported that the cluster threshold was "not detailed in the excerpts". Average answer score: 1.00.

Large chunks (2000) gave the best answers on single-document questions (Q1 and Q2 both 3/3). This was the only configuration in which a whole provision fit into one retrieved chunk. The full Blueprint principle (opt-out, fallback, *and* the domain list) and the whole of EO Sec. 4.2(a), including red-team item (C), arrived intact, so the generator had nothing to miss. But Q3 dropped to retrieval 1. A 2000-character chunk usually spans several topics, so its embedding is a blurred average, and generic, title-heavy introductions (Blueprint p. 4 "About this framework", the p. 2 disclaimer) outranked the specific fallback passage. A second, hidden mechanism makes this worse: **65.8% of the 2000-character chunks exceed MiniLM's 256-token input limit** (and 9.9% exceed mpnet's 384), so the embedder silently truncates them. The retriever ranks a chunk by roughly its first 1100 characters, while the generator receives all 2000. That explains the pattern in Table 5: chunk size 2000 improved the answers much more than the retrieval.

Chunk overlap had almost no effect.

All four overlap settings produced exactly the same scores (Table 5). The chunks did shift: compared with the default, each setting swapped up to three of the four chunks per query, usually for a neighbouring passage of the same section. Overlap 0 and overlap 100 retrieved the same passages for every query (one Q4 chunk differed only in whether it began with "8."). Overlap 400 swapped a duplicated p. 46 chunk on Q1 for a p. 49 passage. None of these swaps added or removed the evidence a question needed. The reasons are structural. Splitting is per page and prefers sentence boundaries, so most cuts already fall between sentences and the overlap only copies whole trailing sentences into the next chunk. More overlap also adds near-duplicates (1,221 chunks at overlap 400

vs 1,009 at 0) that compete for the same top-k slots. Overlap is cheap insurance against a fact being split mid-sentence, but it does not help when the needed sentence sits in the *next* chunk and that chunk does not rank. The Q1 domain list is exactly that case: no overlap setting rescued it.

k traded precision for recall, and the generator benefited from recall.

The answer score rose monotonically with k: 1.40 (k = 2), 1.60 (4), 1.80 (8), 2.00 (12). Retrieval relevance peaked at k = 8 (2.40) and fell at k = 12. For Q2, k = 12 added five clearly irrelevant chunks (an EU article on small-scale providers, the Blueprint's national-security disclaimer, an EO workforce section), so the retrieved set was no longer "highly relevant". The gains came from the long tail. k = 8 pulled in the second half of the Blueprint p. 7 principle at rank 5, finally answering Q1 in full. k = 12 was the **only** MiniLM configuration that reached the EU AI Act on Q3 and Q4 (coverage 0.93): in each case a single EU chunk at rank 11, which was enough to turn a declined comparison into a 2/2 answer. k = 2 was too few: every multi-part question lost a part. More context is not free, though. k = 12 (and k = 2) produced the fabricated "AI system" definition on Q5, and the k = 12 prompts were about three times longer, which costs latency and tokens. For a model like gpt-4o-mini, which handled 12 excerpts well, k around 8 gave the best balance, together with deduplication: identical principle text at p. 7 and p. 46 wasted a slot in almost every Q1 run.

Summary of Experiment B. Chunk size changes *what an embedding represents* and had the largest effect. Too small loses context; too large blurs the vector and gets truncated by the encoder. k changes *how much evidence the generator sees*; a larger k raises answer quality and document coverage at the cost of precision, prompt length and some hallucination risk. Overlap mattered least once the splitter respects sentence boundaries. No parameter setting fixed the core weakness of MiniLM on this corpus, its inability to reach the EU AI Act for cross-document questions, except brute-force k = 12.

4. Conclusion

The single most important lesson: in a multi-document corpus, recall across documents, not the LLM, is the bottleneck, and it has to be engineered and checked deliberately. In 55 of 55 runs the answer was no better than the retrieved evidence. The generator was good at reading broken text (10^{26} from "10 26"), at using a single relevant chunk buried at rank 11, and at admitting gaps. But it cannot synthesize a document that never reaches its context, and with thin context it sometimes filled gaps from its own pre-training. Every major failure in this study was a retrieval failure: MiniLM never reaching the EU Act on Q3 and Q4, the domain sentence just past a chunk boundary, and the duplicated principle text taking up slots. Document-name strings and running headers bias similarity toward documents that repeat their own names. Chunk size and the embedder's token limit interact in ways invisible unless you look at the chunks.

In practice, I would (1) choose the stronger embedder (mpnet here) and keep chunks within its token limit; (2) make cross-document coverage explicit, either by decomposing a comparative or synthesis question into one sub-query per named document, or by metadata-filtered retrieval of the top few chunks *per document*, instead of hoping one global top-k covers all sources; (3) deduplicate near-

identical chunks or use maximal marginal relevance, and strip running headers before embedding; and (4) evaluate by reading the retrieved chunks, as this assignment required. The averages in Table 3 tie on retrieval; only reading the chunks showed that one model was missing an entire document.

Limitations. Five queries and a single human grader on a coarse 3-point scale; scores are judgments, and a different grader could move individual cells by one point. Answers come from a single temperature-0 sample per configuration (cached for reproducibility). Experiment B was run only for MiniLM, as specified.